

	WRITING EFFECTIVE ENGINEERING RESEARCH PAPERS— A BRIEF GUIDE FOR NEW WRITERS	PREPARATION AND EMPLOYMENT OF TEACHERS
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1. INTRODUCTION

One of researchers' most important duties is to share knowledge. When published, a researcher's work has more value and can be used by other professionals. Articles must communicate new and original information, but they must also be written and published to be accessible to international research communities [1]. Along with providing an opportunity for researchers to explain why their work is important, publishing also highlights the advancement in a specific scientific or technological field. Writing effective engineering research papers is not easy and must not be taken lightly. Currently, new writers follow established structures and styles to prepare their first draft. However, engineering research papers are not only rigid documents but also creative intellectual products. Therefore, successful publications require practical and technical writing skills in structuring and phrasing.

Unwritten rules exist about how some sections may vary depending on each particular journal. For example, in the field of engineering, the results and discussion sections are usually combined into one. Other particular strategies may also be followed, but first-time writers need to initially understand what their readers and editors want. Readers and editors interpret information based on the way the text and illustrations are presented. Gopen and Swan [2] suggest always writing "with the reader in mind." In summary, clarity should be the guiding principle for writing all engineering research papers. On the other hand, editors want more than what the readers want. Editors want original engineering research papers which interest and excite readers, contribute to the state of the art, fit within the journal's aims and scope, inspire new ideas and possibilities, and are well written.

Although it may seem easy, the road to publication is almost never straightforward. Vitse and Poland point out that new writers must first have a firm understanding of what should be included in each manuscript [3]. Great papers show innovative results or methods which further the knowledge of a certain field. The researcher's contribution—or main idea—is the core of all engineering research papers; but in practice, beginners find it complicated to write for their target audience. Many books [1, 4, 5] and articles [2, 3, 6-8] explain and provide guidelines on the ways in which scientific papers should be written, organized, and structured, including those that explain techniques to avoid imperfections and those that provide instructions for submission and publishing. However, they rarely provide unwritten rules for achieving success in the field of engineering. The aim of this paper is to focus on simple suggestions, useful tips, and unwritten rules for writing engineering research papers. This article provides a brief guide for new writers.

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2. HOW TO WRITE EFFECTIVE ENGINEERING RESEARCH PAPERS

2.1 WRITING GOOD TITLES

Good titles capture the reader's attention. The purpose of the title is to effectively communicate the content of the article. The more informative the title is, the easier it is for the editors and referees to evaluate how relevant the work is to their potential readers [9]. Although several engineering research papers are published each year, many go almost unnoticed. Currently, many engineers use search engines to quickly search for articles. Letchford et al. [10] found that papers with shorter titles are more frequently cited. Therefore, titles must be accurate and straightforward to attract considerable attention from the readers during a literature search. It is important to remember that a clear title encourages international readers to read—and maybe cite—your paper.

Finding general procedures or basic guidelines to create good titles is possibly complicated. For example, some expert authors recommend writing a certain number of titles and finally choosing the most appropriate one, while others suggest starting with an initial title in mind and so on. Although it might seem simple, choosing a title requires drafting and many hours of preparation. As an unwritten rule, titles should contain some keywords. Draft a title using the fewest possible words—avoiding acronyms, abbreviated words, or terms that are unfamiliar to the readers. Remember that a balance between brevity and verbosity is the key to creating a successful catchy title. Fig. (1) shows some simple suggestions and useful tips for writing good titles.

Title

1. Keep it simple and direct (6 - 12 words)
2. Avoid unnecessary words
3. Use gerundive and infinitive verbal forms when possible
4. Some useful questions:
 - The contribution is featured at the head of the title?
 - What is the main difference?



Fig. 1: Simple suggestions and useful tips for writing good titles

2.2 WRITING WELL-STRUCTURED ABSTRACTS

All engineering research papers have only one opportunity—the abstract—to convince their readers to continue reading. The title attracts the readers, but the abstract explains why the paper is worth reading. It is the most laborious section of an engineering research paper and it should be considered as an independent and self-contained document. The abstract is an abbreviated version—concise and precise—of the article, written in a single paragraph for which every word is chosen carefully. Writing an abstract should not be taken too lightly or as an afterthought. As an unwritten rule, abstracts should contain some marketing communication which tells the readers about the important findings in your article to promote or sell it.

Many common mistakes in abstracts are related to the use of basic structures. Abstracts may be structured or unstructured. Structured abstracts—common in medical and social journals—are written using subheadings which cover the following sections of the paper: background, aim, method, result, and conclusion. Unstructured abstracts—common in engineering journals—contain the same sections but do not explicitly segregate the subheadings [11]. Remember that including all sections of the article accurately and specifically is the key to creating an impactful abstract. Moreover, it is important to avoid abbreviations, symbols, acronyms, trademarks, or unfamiliar terms or referencing other articles, figures, equations, or tables while writing abstracts. Fig. (2) highlights some simple suggestions and useful tips for writing well-structured abstracts.

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Abstract

1. Keep it brief
2. Use present tense
3. Some useful questions:
 - What is the problem to be solved?
 - Why is the problem important?
 - What is the contribution of the paper?
 - What are the key features of the work?
 - Is it clear the level of difficulty of the work?



Fig. 2: Simple suggestions and useful tips for writing well-structured abstracts

2.3 WRITING INTRODUCTIONS

For new authors, the introduction is one of the more complicated sections of the manuscript to write. Editors and referees are the primary reading audience who will decide if the manuscript is worth publishing. Although these concepts might seem obvious, very often, beginners do not write the introduction successfully hence, engineering research papers convey the wrong information to the readers. When drafting an introduction, observe generalized patterns. Swales and Feak [4] have identified a generalized—and flexible—pattern that commonly appear in most research paper introductions: (1) establishing a research field, (2) identifying a research gap, and (3) occupying the niche. It is important to remember that the introduction should highlight the contribution of the paper and keep the reader motivated and interested.

Well-written introductions are constructed as descriptive “funnels”, starting from the general field of study to the specific research topic—or work objectives [7]. Successful writing of introductions starts with identifying the research gap in the existing literature. For an effective introduction, it is necessary to consider the state of the art while explaining why the research is important and to avoid excessive citations; a research paper is not a thesis. The introduction should include approximately three or four paragraphs. Identifying the target audience is also important to decide how extensive the introduction will be. As an unwritten rule, the introduction should convince your readers that your work is relevant and innovative. Fig. (3) lists some simple suggestions and useful tips for writing introductions.

Introduction

1. Use past or present perfect tense to describe the state of the art
2. Use present tense to include the background information
3. Some useful questions:
 - What is the problem?
 - What has been done in the state of the art?
 - What is the contribution of the paper?
 - Why is this research important?
 - Why is this research different?



Fig. 3: Simple suggestions and useful tips for writing introductions

2.4 DESCRIBING MATERIALS AND METHODS

The material and methods section in engineering research articles provides a straightforward description of designs, theories, methods, mathematical models, settings, control strategies, techniques, instruments, vendor names, equipment, and materials used to explain the conducted experiments. The aim of this section is to demonstrate how the research was conducted, providing enough information—or sufficient detail—to ensure reproducibility [12]. Editors and referees will judge the design of your study based on whether another researcher can replicate the experiment or not [8]. Many researchers commonly emphasize on the action using the passive voice in the material and methods section. Many

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authors recommend using active verbs instead of passive verbs to make the writing more direct and less wordy. In any case, only the targeted journal's guidelines for authors have the right answer.

To organize the material and methods section as clearly as possible, the use of subsections is highly recommended. Several researchers divide the subsections while performing experiments to avoid forgetting key details. Generally, methods previously published can be cited without describing them in detail. However, any novel method must be described completely while still avoiding unnecessary details and short subsections. Standard methods, such as analysis of variance (ANOVA), do not require a reference; in this case, naming the method is sufficient. It is necessary to note that the material and methods section is usually best described in the order in which the research was conducted. As an unwritten rule, the material and methods section should be well reported with the use of good figures and tables following a logical order and keeping the explanations brief and concise. Fig. (4) reveals some simple suggestions and useful tips for describing materials and methods.

Material and methods

1. Use past tense to describe what was done
2. Use present tense for diagrams, tables, and figures
3. Use present tense to explain your procedure
4. Do not mix results in this section
5. Do not include unnecessary details
6. Some useful questions:
 - How was the experimental design developed?
 - How can you justify the experimental design chosen?
 - All your constants, variables or parameters are well-defined?
 - What is special, different or unexpected in the experimental design?



Fig. 4: Simple suggestions and useful tips for describing materials and methods

2.5 REPORTING RESULTS AND DISCUSSING FINDINGS

Writing the results and discussion section is like telling a clear and comprehensible story [13]. The aim of this section is to describe and discuss the results in a systematic and detailed way. Referees will surely search for evidence which proves the author's ability to outline and recognize the differences in the results in an objective way. Another purpose of this section is to create cohesion with the material and methods section. Generally, results can be presented either separately from the discussion or together in a single section; this depends mainly on the directions provided by the target journal. Discussing findings is the key to relate the objectives of the work which have been outlined in the introduction section. In the results section, authors only highlight the important findings and describe figures and tables, and in the discussion section, they only explain the results obtained and answer the research question—moving from the particular to the general like an inverted funnel.

For many new writers, the results and discussion section is the starting point of writing a scientific manuscript [7]. This section must be objectively presented; writers should be selective in presenting only the relevant key findings which are relevant to the research question and objectives. Many manuscripts are often rejected because the discussion section is not well-supported by the results. Some common beginners' pitfalls are mainly related to presentation. Do not introduce new information, terms, or ideas not covered in the introduction or the material and methods section. Avoid overstating the significance of the results—overinterpretation may weaken the study. Do not present results that have never been discussed in the text. As an unwritten rule, the results and discussion section must not be monotonous. In this section, you must sell your paper using impactful illustrations. Fig. (5) shows some simple suggestions and useful tips for reporting results and discussing findings.

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Results and discussion

1. Use past tense to describe the results obtained
2. Use present tense for diagrams, tables, and figures
3. Use past tense to summarize findings
4. Use present tense to explain and interpret your results
5. Some useful questions:
 - How was the experimental setup proposed?
 - What initial conditions were selected before testing?
 - Based on the results, what was demonstrated or shown?
 - How can you interpret the significance of the results obtained?
 - Why the results obtained can be considered satisfactory?



Fig. 5: Simple suggestions and useful tips for reporting results and discussing findings

2.6 WRITING CONCLUSIONS

Generally, the conclusion section provides a brief summary—or final opportunity—to explain the significance of the findings based on the results and discussion section [14]. The goal is to clarify how the research question posed in the introduction has been answered in a reader-focused manner, providing the key message and suggesting future research opportunities. This section should not be a copy of the abstract, introduction, or discussion. Remember that a conclusion should be more than a brief summary. A common pitfall when writing the conclusion is the introduction of new statements or arguments that are beyond the scope of the study. It is essential to not include speculation. Fig. (6) lists some simple suggestions and useful tips for writing conclusions.

Conclusion

1. Use a combination of tenses to highlight future directions
2. Do not use more words than necessary
3. Some useful questions:
 - What was demonstrated in this paper?
 - What new possibilities can be obtained with this research?
 - The research promotes new research directions?
 - The conclusion is coherent with the title, abstract, and introduction?



Fig. 6: Simple suggestions and useful tips for writing conclusions

3. HOW TO IMPROVE ENGINEERING RESEARCH PAPERS

3.1 WHY EDITORS AND REFEREES REJECT ENGINEERING RESEARCH PAPERS?

Following the well-known introduction, materials and methods, results and discussion (IMRAD) structure—a standard pattern for writing and publishing many scientific articles in several disciplines—is crucial to prepare effective engineering research papers. Editors-in-chief consider rejecting manuscripts for four reasons: (a) incomplete introduction, (b) insufficient description of methods and materials, (c) overinterpretation of results and discussion, or (d) unclear writing or difficulty in reading the text. Before writing, beginners need to take two important principles into consideration: (a) clear writing requires clear thinking and (b) good writing requires good editing.

Most engineering research papers are published in international journals which use a double-blind peer review process in which referees in the field objectively evaluate scientific work [6]. Through this process, manuscripts can be recommended for publication, revision (minor or major), or rejection. Referees who evaluate manuscripts usually consider the following two main criteria while deciding whether to accept them for publication: (a) the relevance of the contribution and (b) the clarity of the writing style. A clear understanding of these two reasons—some explicit, others

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implied—is crucial when preparing a manuscript for submission. In fact, clear writing is essential to keep the reader motivated and interested in your paper.

3.2 PREPARING FIGURES AND TABLES

Figures and tables are the most efficient way to summarize large amounts of information. Illustrations are critical in transmitting complex ideas concisely in one section. O'Connor and Holmquist [15] state that the writing process starts with drafting figures and tables. The order in which the figures and tables appear in an engineering research paper can help to generate a detailed first draft. Clarity is the key factor for successful illustrations. Each figure and table should have a short caption that clearly explains the illustrated information. Their captions should be able to stand alone and be self-explanatory.

Pictures can help in reducing the length of the text and increasing the visual impact of your engineering research paper. Editors and referees can interpret results more easily if the figures and tables are formatted using graphic design software tools so that the data can be better presented. Most international journals suggest specific formats for their articles, such as the type of file, file size, and resolution. Quite often, each guide for authors explains how to submit figures and tables, the format and the quality of each illustration [5]. When preparing the manuscript, it is important to ensure that all figures and tables are numbered and cited in the text. As an unwritten rule, take professional photographs using a high-resolution camera. It is essential to note that the appearance of your illustrations affects the reception of your article.

3.3 REFERRING TO LITERATURE

A good references section is an essential part of a scholarly work. References allow the readers to authenticate the information and statements presented in the article. An important factor to remember while writing the references section is to wisely choose quality over quantity so, you should include only essential references and eliminate those that are irrelevant or archaic—a good reference list has recent literature no older than four years. When citing the references, write them correctly, appropriately citing all the articles. It is necessary to avoid excessive self-citations and not include articles which have not been published in English or not yet accepted for publication. To simplify electronic access, nowadays, many articles have a digital object identifier (DOI) that can identify them permanently and unambiguously. As an unwritten rule, use 10–20, 25–40, and 50–80 references for conferences, researches, and review papers, respectively.

There are two main referencing styles widely used in articles and books—APA and Vancouver. Each style differs mainly in the manner that the references are presented. For example, APA style uses an author-date citation while Vancouver uses Arabic numerals. It is important to check the style of the references section before submitting the manuscript. If the authors—especially beginners—do not follow the required style by the journal, editors or referees can reject the submitted manuscript. Making errors when writing the references section is very common. EndNote is one of the most widely used reference software tool for grouping, managing, and formatting bibliographies. Many researchers use EndNote to change the referencing format and style according to the target journal. However, you must be cautious with EndNote as sometimes the bibliographic style does not format correctly, thus appears differently in the word processor program.

3.4 WRITING ACKNOWLEDGMENTS

The acknowledgments section is written after the conclusion section. Its purpose is to thank all those who contributed to the research but cannot be listed as authors. For example, people who provide technical or intellectual assistance can be acknowledged in this section. Moreover, you can use this section to provide information about the funding agencies that contributed to the realization of your research—some international journals request that the acronym and the grant number be inserted. When thanking financial resources or persons, it is important to write their names in their native language.

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3.5 AVOIDING PLAGIARISM

Plagiarism is a serious problem in academia. Currently, many international journals use plagiarism detection software, such as Turnitin to detect potential plagiarism. Nevertheless, editors and reviewers still find identical articles, duplicate illustrations, or similar sections during manuscript revision. As a consequence, researchers may be subject to personal, professional, ethical, legal, and monetary repercussions. Self-plagiarizing is the reuse of previously published papers without citing one's own work. Generally, some researchers make the mistake of copying their own methods without divulging that their work has been previously published. Furthermore, authors should avoid salami publication. It is important to not practice plagiarism or self-plagiarizing—be professional and cite others' work properly.

4. CONCLUSION

This paper offers a brief guide that may help new writers prepare effective engineering research papers. The aim of this paper was to focus on simple suggestions, useful tips, and unwritten rules for writing engineering research papers. Acquiring writing skills to produce well-written engineering articles will surely increase the likelihood that a submitted manuscript will be successfully published. Remember that writing engineering research papers that get published goes beyond manuscript preparation—it is an art and a craft. Authors must explain the information in a clear and concise way—always with the reader in mind; this is the key contribution of the researcher. Finally, manuscript preparation becomes considerably simpler when the research question—or contribution—is clearly defined. Therefore, the key to successful publishing is answering this question correctly.

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